

Bachelor's Thesis

Quantifying Labelling Uncertainty in Image Data

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Abstract

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1 Data

1.1 Introduction to the Datasets

We will analyse labelling uncertainty in the context of image classification using the *CIFAR-10H* dataset. The dataset is based on the *CIFAR-10* (Canadian Institute for Advanced Research, 10 classes) dataset, first introduced in Krizhevsky (2009), which is a benchmark dataset for image classification and is itself a subset of the larger *80 million tiny images* dataset, see Torralba et al. (2008). The *CIFAR-10* dataset consists of 60'000 images of size 32×32 pixels. Each image belongs to one of ten classes: airplane, automobile, bird, cat, deer, dog, frog, horse, ship, and truck. Some example images of each class are shown in Figure 1. The images in the *CIFAR-10* dataset come with multiple advantages for our purposes, see Peterson et al. (2019). First, the dataset is well-known and widely used in the machine learning community, see *Papers with Code - CIFAR-10 Dataset* (2024) for a list of papers that use the dataset. Second, the low resolution of the images makes the dataset challenging for image classification and introduces ambiguity in the labelling process. Third, the dataset contains images that are close to the category boundaries, which further introduces ambiguity in the labelling process. This is in contrast with other datasets that are more carefully created such that each image is selected to be a unambiguously of a certain category.

The *CIFAR-10* dataset is divided into a training set of 50'000 images and a test set of 10'000 images. The test set is the basis for the *CIFAR-10H* dataset, which will be used in this thesis. *CIFAR-10H* is a balanced dataset, meaning that each class contains the same number of images, i.e. 1000 images. *CIFAR-10H* is a human-labelled dataset, where each image has been labelled by multiple human annotators. The dataset contains 514'200 observations from a total of 2571 annotators. Each annotator labelled 200 images, 20 from each category, which were randomly sampled from the test set of the *CIFAR-10* dataset. On average, each image was labelled 51 times, with the minimum number of labels being 50 and the maximum number of labels being 63. The relevant features of the *CIFAR-10H* dataset are described in Table 1. The study design is described in more detail in Peterson et al. (2019), section 4.2. Note, that the images were collected based on searching the web, using search engines. Thus, the true label is not necessarily the object that is displayed in the image, but the term that was used to search for the image, see Torralba et al. (2008). For our purposes, we will assume that the true label is the object that is displayed in the image.

The *CIFAR-10* and *CIFAR-10H* datasets are publicly available and can be downloaded from the CIFAR-10 website *CIFAR-10 and CIFAR-100 Datasets* (n.d.) and the author's GitHub repository *Jcpeterson/Cifar-10h: Original Dataset Release for CIFAR-10H* (n.d.).

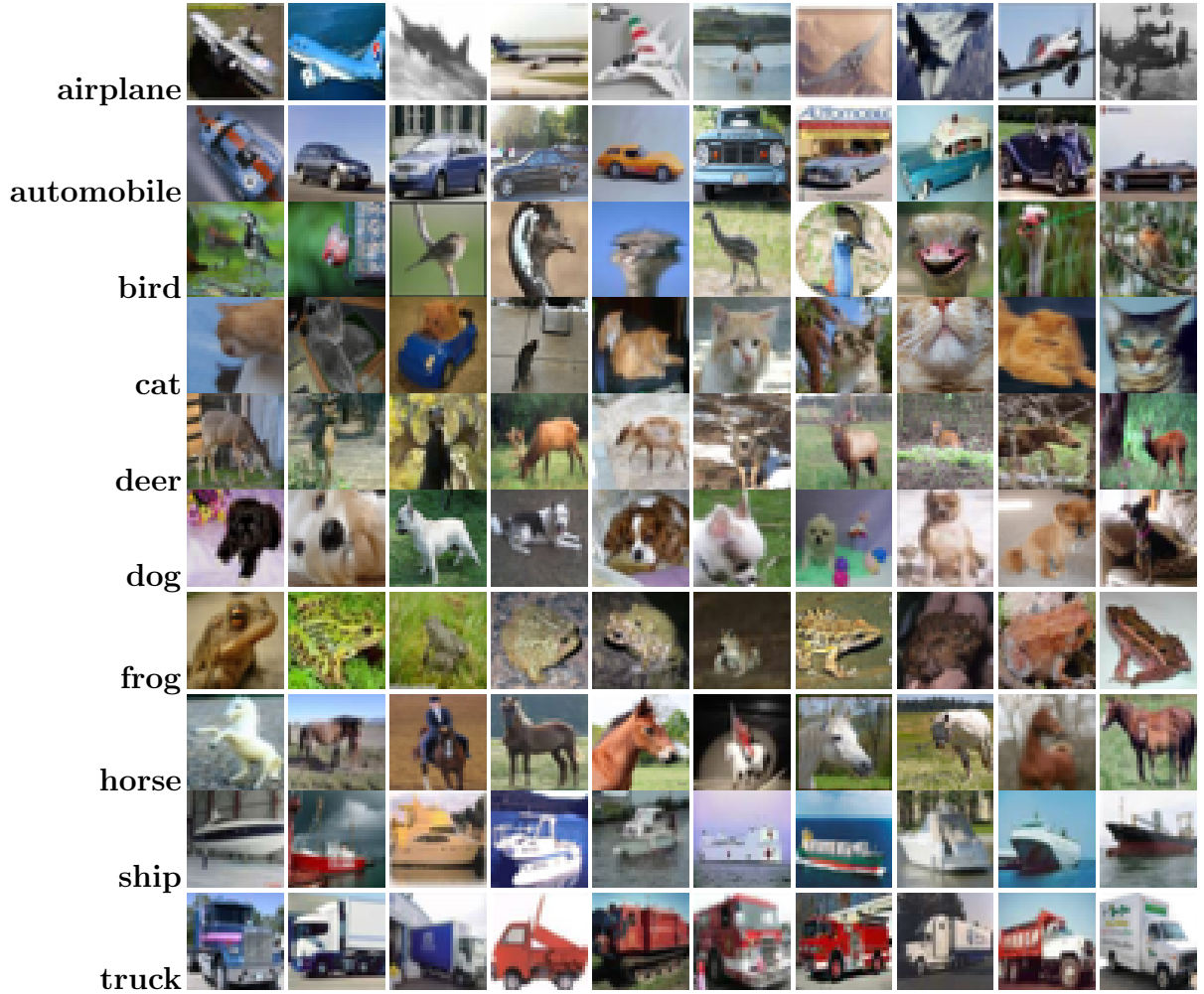


Figure 1: The classes of the CIFAR-10 dataset with ten example images per class.

Feature	Description
annotator_id	Unique integer ID ranging from 0 to 2570
trial_index	Indexes the trial for each annotator, from 0 to 219
true_category	True string class name (e.g., "cat") of the image. The set of categories is {airplane, automobile, bird, cat, deer, dog, frog, horse, ship, truck}
chosen_category	String class name (e.g., "cat") chosen by the annotator for the image from the set of possible true categories
correct_guess	Whether the guess was correct (i.e., when true_category == chosen_category), marked by 1, or 0 otherwise
image_filename	Filename of the PNG (e.g., "cabin_cruiser_s_000814.png")
subcategory	Subordinate class label, which is recoverable from the image filenames (e.g., "cabin_cruiser")
reaction_time	Time taken (in milliseconds) by the annotator to provide a response for the current trial
time_elapsed	Total time (in milliseconds) that passed at the end of each trial. The first is equal to the reaction_time for the first trial. The rest are essentially cumulative reaction_time values

Table 1: Features of the CIFAR-10H dataset.

1.2 Descriptive Statistics

As mentioned in the previous section, the *CIFAR-10H* dataset is balanced with respect to the true categories. This means, that we have 51'420 observations per true category. But the number of observations per chosen category is not balanced, see Figure 2, e.g. the category *deer* is underrepresented, while the category *dog* is overrepresented. The question that arises is, whether chosen categories are usually correct or not and where the votes of overrepresented categories come from.

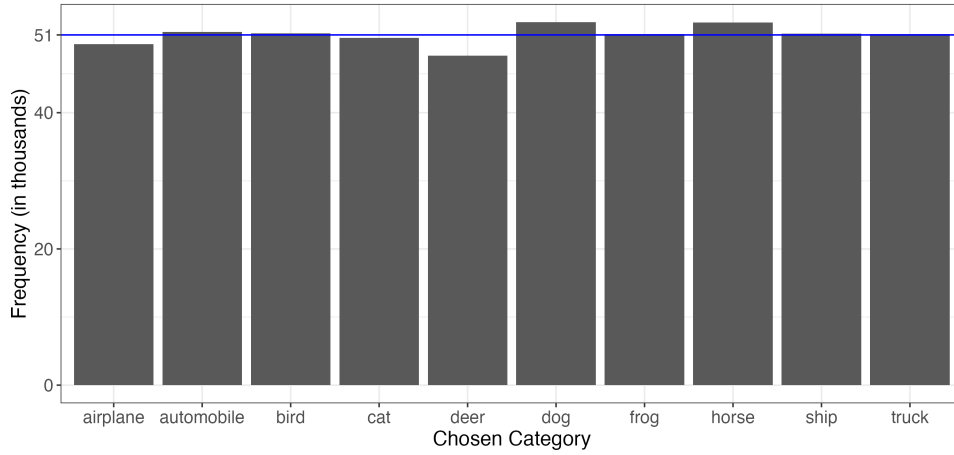


Figure 2: Number of observations per chosen category. The blue horizontal line indicates the 51'420 observations per true category, so that we can see, which categories are over- and underrepresented.

We can also question the ambiguity of the images. For that, we can check, how many unique labels each image has received. If an image is unambiguous, then all annotators should agree on the label of the image. If an image is ambiguous, then there are multiple possible labels for the image. In Figure 3, we can see the distribution of the number of labels per image for each true category. We can see, that in all categories, 50% of the images have received at most 2 labels. We can also see, that in almost all categories, 25% of the images have received 3 labels or more. This indicates that a lot of images are unambiguous, but also that a considerable amount of images are ambiguous.

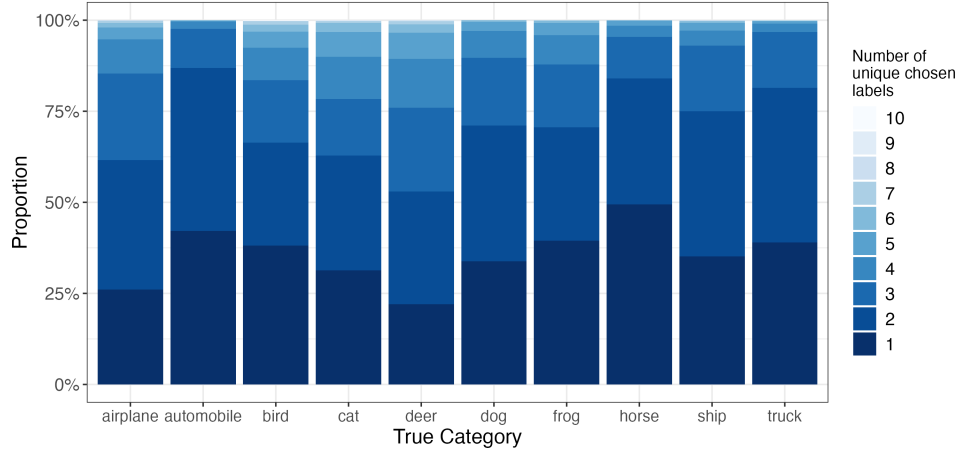


Figure 3: Distribution of the number of unique chosen labels per image for each true category. Unambiguous images have received only one label, while ambiguous images have received multiple labels. The more labels an image has received, the more ambiguous the image is.

In Figure 4, we can see an examples of ambiguous and unambiguous images as well as their respective vote distribution.

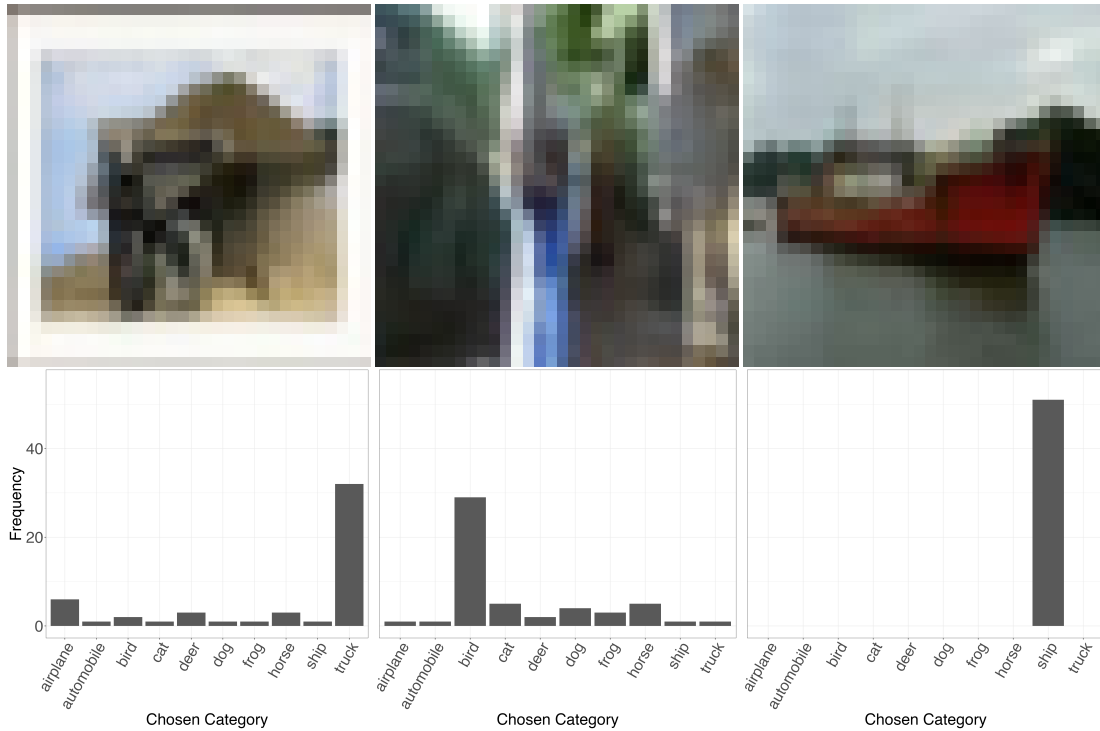


Figure 4: Example of an ambiguous image. The image shows a car, but the chosen categories are *automobile* and *truck*.

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